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Potential is “intrinsic energy of inertia”

elec. Potential structure:

$$U_e := \Phi \stackrel{\text{"ignition, cause, inhomogeneity", source at rest, } v=0}{=} -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\epsilon_0} \int \frac{\varrho}{|r|} dr' \stackrel{=}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\epsilon_0} Q = \quad (1)$$

magn. Potential structure:

$$U_m := A \stackrel{\text{"ignition, cause, inhomogeneity", source at rest, } v=0}{=} -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \stackrel{=}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \quad (2)$$

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (3)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (4)$$

Electric induced energy $E_{kin} = \frac{1}{2}mv^2 \stackrel{=}{=} (pc)^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \epsilon_0 \mathcal{E}^2 \quad (5)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (6)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (7)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (8)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (9)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \quad (10)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (11)$$

Energy conservation for (large extended) conductive medium if no forces act on it:

$$F = \int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\right) \quad (12)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (13)$$

V = Potentialdifference I = Current R = Resistance j = Current density J = "ignition of inhomogeneity, c

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector = radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor}$ $r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$

$\rho = kr = \text{"radius of curvature } k"$ $\omega = 2\pi\nu = k\lambda\nu = \frac{1}{\sqrt{LC}} = \text{frequency}$

μ = permeability of conductor σ = conductivity of conductor

Low Frequency:

$$\begin{aligned} \omega &\ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \\ L &\approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0 \end{aligned} \quad (14)$$

High Frequency:

$$\begin{aligned} \omega &\gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \\ L(t) &\neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0 \end{aligned} \quad (15)$$

Energy:

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\alpha) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{aligned} \quad (16)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (17)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (18)$$

V_0 = Opposing Potential ($I = 0$)

$\alpha = \angle(E_{kin}, \nu)$ e = electron charge Q = ion charge

¹Wirkungsquantum

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element